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Research Article

Dual-Layer Grain-Boundary In Situ Polymerization Modulates Elastic Modulus for Mechanically Stable Flexible All-Perovskite Tandem Solar Cells

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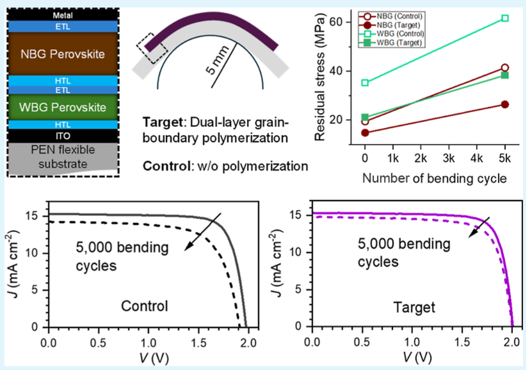

ABSTRACT: Flexible all-perovskite tandem solar cells (PTSCs) are promising for lightweight photovoltaics, yet their mechanical degradation under bending remains poorly understood. Although bending instability has often been attributed to brittle fractures, halide perovskites are mechanically soft materials. Here, we demonstrate that bending-induced degradation in flexible PTSCs primarily associated with strain-induced evolution of residual-stress accumulation prior to crack formation, while elastic-modulus engineering through grain-boundary polymerization mitigates this mechanically induced degradation. Grazing-incidence X-ray diffraction analyses reveal that repeated bending accumulates residual stress of both wide-bandgap and narrow-bandgap perovskite layers in PTSCs, leading to fatigue-associated degradation and enhanced nonradiative recombination even in the absence of visible cracks. To address this intrinsic limitation, a dual-layer grain-boundary in situ polymerization strategy is introduced for both perovskite layers, enabling elastic relaxation and suppressing residual-stress accumulation. As a result, flexible PTSCs achieve a power conversion efficiency of 24.97% and retain over 90% of their initial efficiency after 5000 bending cycles at a radius of 5 mm. This work establishes elastic-modulus engineering as a key design principle for mechanically robust flexible perovskite tandem solar cells.

KEYWORDS: *all perovskite tandem, flexible solar cell, bending durability, in situ polymerization, grain-boundary passivation*

1. INTRODUCTION

Perovskite solar cells (PSCs) have emerged as leading candidates for high-efficiency, low-cost photovoltaics following the seminal 2012 report on solid-state PSCs,[1] owing to the unprecedented optoelectronic properties of organic lead iodide perovskite absorbers. As a result, the certified power conversion efficiency (PCE) exceeds 27% in single-junction devices and approaches 35% in dual-junction tandem architecture.[2−4] The tandem architecture can be realized by integrating a wide bandgap (WBG) perovskite top subcell with a narrow bandgap (NBG) bottom subcell comprising silicon, CIGS, Sn−Pb perovskite or organic materials. Among these architectures, PTSCs represent a particularly promising technology, owing to their lightweight form factor and high efficiency enabled by solution-processable fabrication.[5,6] Flexible PTSCs further expand their application scope to wearable electronics and aerospace systems due to their mechanical bendability and low-cost, lightweight device architecture.[7,8] However, despite these advantages, their mechanical robustness under realistic conditions remains a persistent challenge.[9] The heterojunctional tandem structure inherently comprises multiple functional layers with diverse mechanical and thermal properties, including elastic modulus (E) and coefficient of thermal expansion (CTE).[10−12] Mismatches in these properties lead to the accumulation of residual stress during fabrication, which readily triggers mechanical fatigue, grain boundary cracking, and delamination, ultimately compromising both device efficiency and long-term stability.[9,13] Various strategies have been proposed to enhance the mechanical stability of flexible PTSCs, including nano-composite interfaces, ductile electrode materials, and grain orientation control.[10,14,15] While diverse studies have investigated strain modulation in perovskite films through additive engineering and interfacial tuning, reports specifically addressing strain in perovskite layers within flexible tandem architectures remain relatively underexplored. A promising approach is the incorporation of polymeric interphases that

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provide mechanical elasticity, effective defect passivation, and enhanced resistance to environmental stress.[16] Particularly, in situ polymerization at grain boundaries of perovskite films is an effective route to enhance simultaneous passivation and mechanical robustness.[17−19]

Here, we report a dual-layer grain-boundary polymerization strategy to enhance bendability and mechanical stability in flexible two-terminal (2T) PTSCs using bisphenol A diglycidyl ether (DGEBA) and isophorone diamine (IPDA). Since we previously found that the reaction between the epoxy resin DGEBA and the cross-linking agent IPDA led to in situ polymer formation at grain boundaries,[20] this approach is expected to improve the elastic properties of perovskite thin films by reducing their *E*s, thereby enhancing their ability to withstand mechanical stress in flexible PTSCs. Grazing-incidence X-ray diffraction (GIXRD) analyses confirm a reduction in residual stress generated during the formation of both NBG and WBG perovskite films, as well as a suppression of stress-fatigue accumulation under repetitive bending strain. Suppression of mechanical deformation in both interfacial and bulk layers and nonradiative recombination traps are realized by the dual-layer grain-boundary polymerization, leading to a best PCE of 24.97%, together with an improved durability, maintaining over 90% of initial PCE after 5000 times of repetitive bending cycles at a bending radius of 5 mm.

2. MATERIALS AND METHODS

2.1. Materials

Formamidinium iodide (FAI; FA = $HC(NH_2)_2^+$) and methylammonium iodide (MAI; MA = $CH_3NH_3^+$) were synthesized by reacting 30 mL of hydroiodic acid (57 wt % in water, Sigma-Aldrich) with 23.6 g of formamidinium acetate (99%, Sigma-Aldrich) and 18.2 mL of methylamine (40 wt % in methanol, TCI) in an ice bath. After stirring for 30 min, the resulting dark precipitate was isolated by solvent evaporation at 60 °C using a rotary evaporator. The crude solids were washed multiple times with diethyl ether (99.0%, Samchun), recrystallized from anhydrous ethanol, and vacuum-dried for 48 h, which was stored in an argon-filled glovebox. Anhydrous solvents such as dimethylformamide (DMF, 99.8%), dimethyl sulfoxide (DMSO, ≥99.9%), toluene (99.8%), chlorobenzene (CB, 99.8%), and methanol (≥99.9%) were used as received from Sigma-Aldrich. Other materials, including 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), tin(II) iodide (SnI_2, 99.99%), tin(II) fluoride (SnF_2, 99%), thiourea (≥99.999%, metal basis), and phenylhydrazine (PH, 97%), aluminum oxide nanoparticle solutions (Al_2O_3), bisphenol A diglycidyl ether (DGEBA, average M_n 377), isophorone diamine (IPDA, ≥99%) were purchased from Sigma-Aldrich. Lead iodide (PbI_2, 99.99%, trace metals basis), piperazine dihydriodide (>98%), [4-(3,6-Dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) were purchased from TCI. PEDOT:PSS (Al 4083, 1.5 wt % in water) was purchased from Heraeus Clevios and C_{60} was supplied by Nano-C.

2.2. Device Fabrication

2.2.1. Fabrication of Flexible WBG PSCs. Flexible WBG PSCs were fabricated on a patterned indium tin oxide (ITO) deposited on a 50 μm-thick polyethylene naphthalate (PEN) substrate, where the ITO/PEN substrate was attached on a polydimethylsiloxane (PDMS)-coated glass to improve coating quality. The PDMS-coated glass was removed after coating. Me-4PACz layer was first formed on the O_2 plasma-treated substrate by spin coating 1 mM Me-4PACz solution in IPA at 4000 rpm for 20 s and annealing at 110 °C for 10 min. To improve wettability of the perovskite solution, 5.0 mg mL^{-1} of aluminum oxide nanoparticle solution was spin coated on the Me-4PACz layer at 4000 rpm for 30 s, followed by annealing at 110 °C for 10 min. An 1.2 M $FA_{0.8}Cs_{0.2}Pb(I_{0.63}Br_{0.37})_3$ perovskite precursor dissolved in DMSO/DMF mixed solvents (DMSO/DMF = 1:4, v/v) was spin coated at 1000 rpm for 5 s and consecutively 5000 rpm for 30 s. DGEBA was added in the perovskite precursor solution (0.1 mg mL^{-1}) and IPDA was added to the CB antisolvent (1×10^{-3} mg mL^{-1}) for in situ polymerization for target devices. 130 μL CB was dropped on the film at 15 s before end of the spinning and the intermediate phase film was annealed at 110 °C for 50 min. 2 mM piperazine dihydriodide solution in IPA was post-treated on the perovskite film by spin coating and heat-treatment at 100 °C for 3 min. C60 (∼25 nm), BCP (∼6 nm), and 100 nm Ag (∼100 nm) layers were sequentially deposited by thermal evaporation under a vacuum of $<10^{-6}$ Torr. All spin coating processes were proceeded in N_2 filled glovebox and all solutions for spin coating were filtered with 0.2 μm PTFE filters.

2.2.2. Fabrication of Flexible NBG PSCs. A patterned ITO deposited on a 50 μm-thick PEN substrate was used for flexible NBG PSCs, where a PDMS-coated glass was also used to improve coating quality. An aqueous PEDOT:PSS solution mixed with methanol (1:1 by volume) was filtered through a 0.45 μm PVDF filter, which was spin-coated on a O_2 plasma-treated substrate at 4000 rpm for 30 s and then annealed at 120 °C for 10 min in ambient air. The PEDOT:PSS-coated substrates were transferred into a N_2 filled glovebox (O_2 and H_2O < 1.5 ppm). An 1.8 M solution of $FA_{0.5}MA_{0.5}Pb_{0.5}Sn_{0.5}I_3$ was prepared by dissolving 230.5 mg of PbI_2, 186.25 mg of SnI_2, 86 mg of FAI, 79.5 mg of MAI, 7.83 mg of SnF_2, and 6 mol % thiourea in a solvent mixture of 444 μL DMF and 111 μL DMSO.[21] DGEBA was added in the perovskite precursor solution (0.1 mg mL^{-1}) and IPDA was added to the CB antisolvent (1×10^{-3} mg mL^{-1}) for in situ polymerization for target devices. The solution was filtered through a 0.20 μm PTFE membrane before use. The precursor solution was spin-coated onto the PEDOT:PSS layer at 1000 rpm for 10 s and at 4000 rpm for 40 s. During spinning, 300 μL of CB was dropped onto the film 20 s before completion of spin coating. Films were then annealed at 100 °C for 10 min. After cooling, a post-treatment was performed by spin-coating 70 μL of a 52 mM PH solution dissolved in CB at 5000 rpm for 30 s, which was followed by heating at 65 °C for 5 min[22] Finally, C_{60} (∼20 nm, 0.2 Å/s), BCP (∼6 nm, 0.2 Å/s), and Cu (∼100 nm, 0.3 Å/s) were thermally evaporated sequentially under high vacuum (pressure $<10^{-6}$ Torr).

2.2.3. Fabrication of Flexible 2T PTSCs. The flexible WBG perovskite subcells were fabricated by the aforementioned procedure. After the deposition of C_{60} (∼20 nm) on top of the WBG perovskite layer, the substrates were transferred to an atomic layer deposition (ALD) system to deposit a 30 nm-thick SnO_2 film at 85 °C. The substrates were then transferred back to a thermal evaporator to deposit an ultrathin Au layer (∼1 nm) on the ALD-SnO_2 layer. Subsequently, a PEDOT:PSS film was formed on the Au layer, which was followed by the same fabrication process used for the aforementioned NBG subcells.

2.3. Characterization and Measurements

J−*V* characteristics and maximum power point tracking (MPPT) were measured under AM 1.5G illumination (100 mW cm^{-2}) using a solar simulator (Oriel Sol 3A) and a Keithley 2400 source meter. The light intensity was calibrated with an NREL-certified silicon reference cell equipped with a KG-5 filter for WBG and a KG-2 filter for NBG and tandem perovskite solar cells. Metal shadow masks with aperture areas of 0.125 cm^2 and 0.100 cm^2 were used for single-junction and tandem devices, respectively. Box plots of the photovoltaic parameters for the single-junction and tandem devices represent the statistical distributions of seven individual devices under each corresponding condition. Representative grazing incident X-ray diffraction (GIXRD) patterns for flexible tandem and single-junction samples were measured using either A8 Discover (Bruker) or SmartLab (Rigaku) with Cu Kα radiation (λ = 1.5406 Å). Tilting angles of 0, 16.78, 24.10, 30.00, 35.26, 40.20, and 45.00° were used. For tandem-structured samples, an incident angle of 7° was employed. Representative nanoindentation measurements were conducted measured using atomic force microscopy (AFM, Park Systems, NX10) under nitrogen

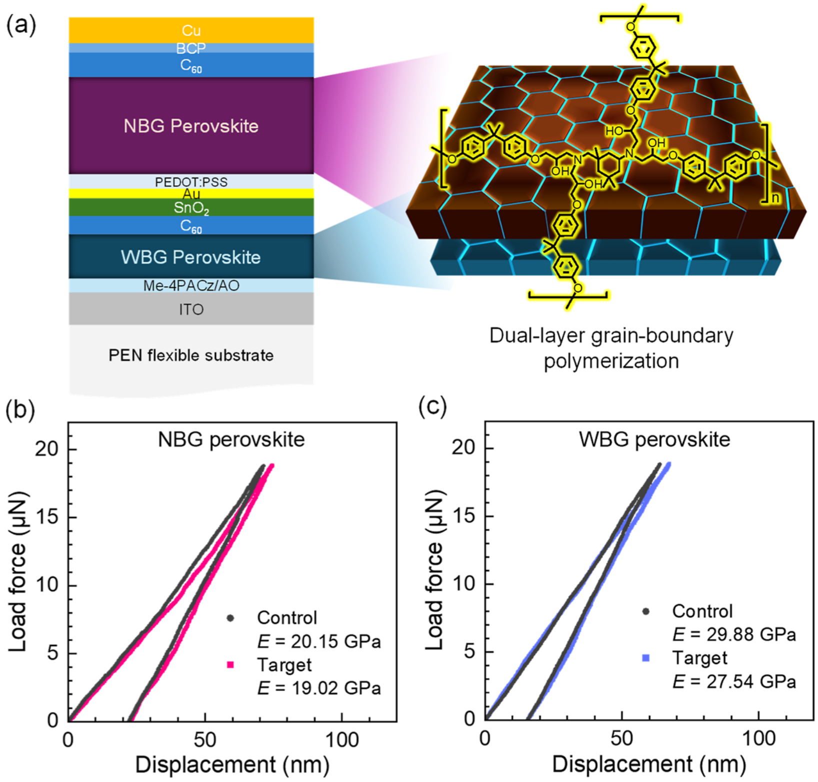

Figure 1. (a) Schematics of a dual-layer grain-boundary polymerization-applied flexible 2T PTSC structure. Load−displacement curves obtained from AFM-based nanoindentation measurements for (b) NBG and (c) WBG perovskite films deposited on ITO/PEN flexible substrate.

atmosphere using Berkovich-type indenter tip (Bruker, NM-RC-SEM), which has elastic modulus (E_i) of 1140 GPa, Poisson ratio (ν_i) of 0.07 and geometry correction factor (β) of 1.034. The AFM tip sensitivity and force−distance slope were calibrated using a SiO_2 calibration standard (HS-100MG, BudgetSensors). Nanoindentation measurements were carried out in a piezo voltage-controlled mode using a voltage limit of 3 V and a loading rate of 0.3 V s^{-1}. The Poisson's ratio of perovskite films ($\nu = 0.33$) used for the derivation of residual stress and elastic modulus was adopted from the literature.[23] The external quantum efficiency (EQE) was measured using a QUANTX-300 QE measurement (Newport) equipped with a 100 W xenon lamp source and 130 mm focal length monochromator with dual gratings. EQE measurements were performed in ambient air and a chopper frequency of 25 Hz. For tandem solar cells, the bias illumination from NIR LED illuminator with emission peak of 850 nm and blue illuminator with emission peak of 460 nm were used for the measurements of the WBG front cell and NBG rear cells, respectively. Bending stability was evaluated using a custom-designed automated bending system that repeatedly applied bending cycles with a radius of 5 mm. Each bending cycle was performed at 1 s intervals, and all tests were conducted in a nitrogen-filled glovebox. Steady-state Photoluminescence (SSPL) measurements were performed using an FLS-1000 spectrometer (Edinburgh Instruments) with excitation provided by a 450 W xenon arc lamp. Time-resolved Photoluminescence (TRPL) was measured using 405 nm pulsed diode lasers (EPL, Edinburgh Instruments). Electrochemical impedance spectroscopy (EIS) measurements were performed using an Autolab 302B (Metrohm) under 1 sun illumination with a 0.9 V bias. Cross-sectional scanning electron microscopy (SEM) images of the flexible tandem devices before and after bending were obtained from each sample using a Helios 5 UX (Thermo Fisher Scientific) system after focused ion beam (FIB) milling. Scanning transmission electron microscopy−energy dispersive X-ray spectroscopy (STEM−EDS) measurements were conducted using a high-resolution transmission electron microscope (HRTEM, JEOL JEM-ARM200F).

3. RESULTS AND DISCUSSION

Flexible 2T PTSCs were fabricated on a 50 μm-thick PEN flexible substrate, as illustrated in Figure 1a. The device structure consists of a bottom subcell comprising an NBG perovskite $FA_{0.5}MA_{0.5}Sn_{0.5}Pb_{0.5}I_3$ with a bandgap (E_g) of 1.25 eV and a top subcell employing a WBG perovskite $FA_{0.8}Cs_{0.2}Pb(I_{0.63}Br_{0.37})_3$ with an E_g of 1.78 eV. Two subcells are interconnected through a SnO_2 layer formed by ALD method and a 1 nm-thick Au recombination layer. As illustrated in Figure S1, in situ epoxy polymerization occurs at the grain boundaries through the reaction of predeposited DGEBA with IPDA during the antisolvent dripping step and the subsequent annealing process at 100 °C.[20] STEM-EDS line-scan analysis reveals a more than 2-fold increase in the carbon signal at the grain boundaries of both NBG and WBG perovskite layers in the flexible 2T PTSC, accompanied by reduced Pb and I signals, supporting preferential localization of the formed polymer at the grain boundaries (Figure S2). Compared to the control without in situ polymerization, the grain-boundary polymerization (hereafter designated as target) improves photovoltaic parameters for both NBG and WBG perovskite films (Figure S3 and Table S1). Load−displacement curves of NBG and WBG perovskite films on flexible ITO/PEN substrates are obtained via AFM-based nanoindentation in Figure 1b,c to estimate E of perovskite films using Oliver-Pharr model as described in eq 1[24]

$$\frac{2\beta\sqrt{A}}{\sqrt{\pi}\,S} = \frac{E}{(1-\nu^2)} + \frac{E_i}{(1-\nu_i^2)} \quad (1)$$

where S represents the unloading stiffness, A is the projected contact area of the indenter, β is the geometry correction factor, ν is the Poisson's ratio of the perovskite, ν_i is the

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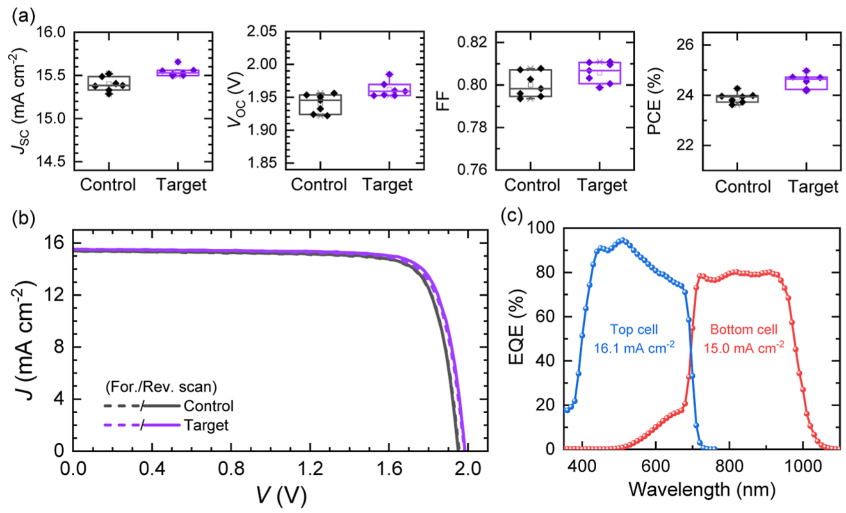

Figure 2. (a) Statistical photovoltaic parameters of J_{SC}, V_{OC}, FF and PCE for control and target flexible PTSCs. (b) $J-V$ curves of champion control and target flexible PTSCs. (c) EQE spectra of the target flexible PTSC.

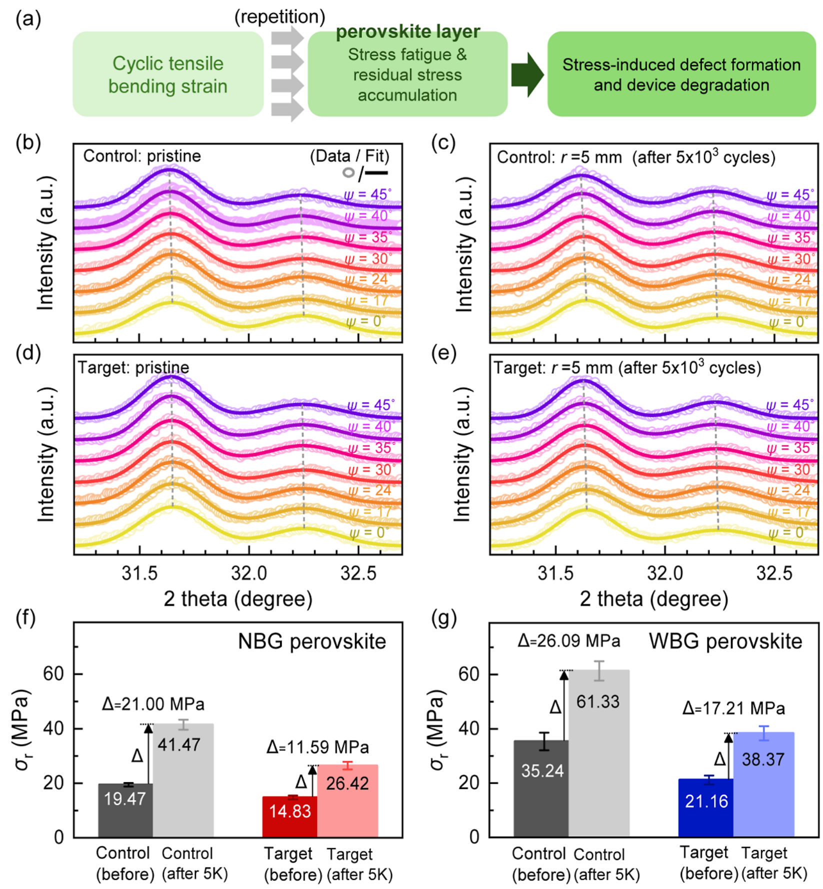

Figure 3. (a) Schematic illustration of the evolution of effects in the perovskite layer and flexible device induced by cyclic tensile bending strain. (b−e) GIXRD patterns of the flexible tandem samples without the rear electrode (Cu), having the structure of PEN/ITO/Me-4PACz/AO/WBG perovskite/C_{60}/SnO_2/Au/PEDOT:PSS/NBG perovskite/C_{60}/BCP structure (b, d) before and (c, e) after 5000 cycles of repetitive bending (bending radius = 5 mm) for (b, c) control and (d, e) target samples. The σ_r before and after 5000 bending cycles estimated based on 2θ-$\sin^2\psi$ relation for the (f) NBG and (g) WBG perovskite layers in tandem-structured samples. The error bars indicate 95% confidence interval of the data.

Poisson's ratio of the indenter, E is elastic modulus of the perovskite, and E_i is elastic modulus of the indenter. Detailed measurement conditions are described in the Experimental section. E of the control perovskite films decreases from 20.15 to 19.02 GPa for NBG and from 29.88 to 27.54 GPa for WBG after in situ polymerization. A similar tendency is observed for the rigid substrate (Figure S4), indicating that the reduction in E originates from the polymerization effect rather than substrate-induced artifacts. The measured elastic modulus reflects the effect of grain-boundary polymerization and the intrinsic mechanical characteristics of the as-formed films. This decrease in E implies an enhanced elasticity for both NBG and

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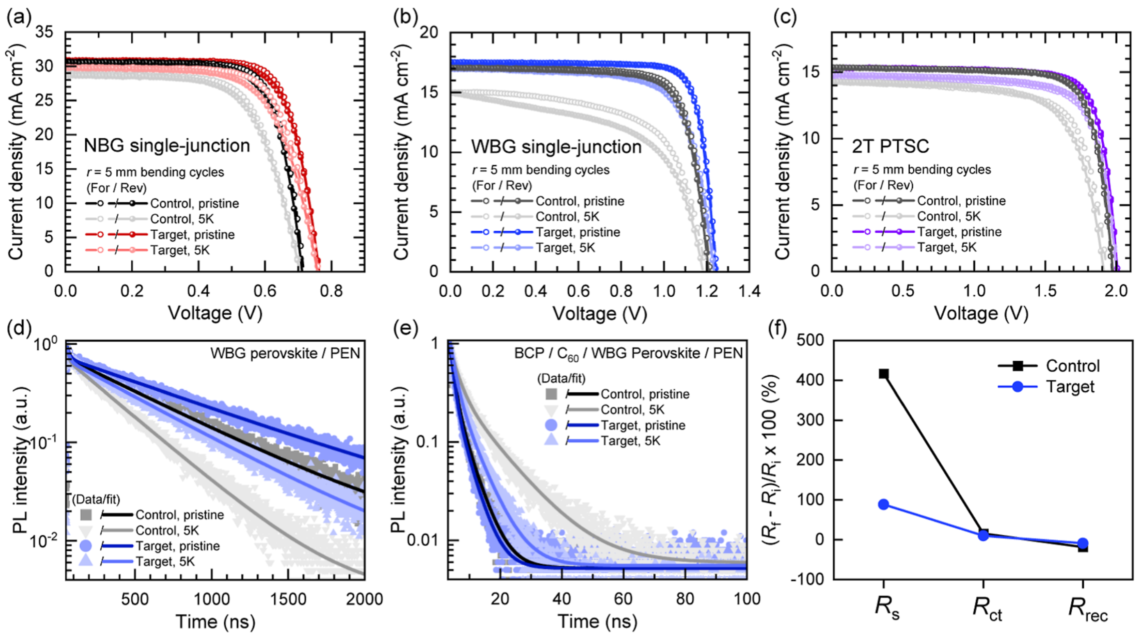

Figure 4. Mechanical durability tracking results of control and target devices and samples. $J-V$ characteristics of the single-junction (a) NBG, (b) WBG cells, and (c) PTSCs under bending strain. TRPL decay of (d) WBG perovskite/PEN and (e) BCP/C_{60}/WBG perovskite/PEN flexible samples. (f) The relative resistance changes of the single-junction WBG cells under bending strain derived from EIS Nyquist plot. R_i is the initial resistance without bending and R_f is the resistance after 5000 bending cycles. All measurements were conducted both in the pristine state and after 5000 bending cycles at the bending radius of 5 mm.

WBG perovskite films, which is expected to be beneficial for accommodating mechanical deformation in flexible devices.[25,26] Excessive polymerization may further improve elastic properties of the perovskite layer, while it can hinder charge extraction by forming thick insulating polymer layers at grain boundaries.[20]

The dual-layer grain-boundary in situ polymerization was applied to a flexible PTSC. As shown in the statistical data in Figure 2a, the average short-circuit current density (J_{SC}) is improved from 15.40 ± 0.08 mA cm^{-2} (control) to 15.54 ± 0.05 mA cm^{-2} (target). Additionally, the average open-circuit voltage (V_{OC}) and fill factor (FF) are also increased from 1.941 ± 0.013 V and 0.800 ± 0.005 to 1.962 ± 0.011 V and 0.806 ± 0.005, respectively. Consequently, the target devices exhibit an average PCE of 24.57 ± 0.26%, which is higher than that of the control devices (23.90 ± 0.19%), as listed in Table S2. Figure 2b presents the current density−voltage ($J-V$) curves of the best-performing PTSCs with and without the dual-layer grain boundary polymerization, where the associated photovoltaic parameters are listed in Table S3. The control device delivers a J_{SC} of 15.41 mA cm^{-2}, an V_{OC} of 1.951 V, and a FF of 0.808, corresponding to a reverse-scanned PCE of 24.27%. The application of in situ polymerization improves PCE to 24.97% (J_{SC} = 15.51 mA cm^{-2}, V_{OC} = 1.985 V and FF = 0.811). Furthermore, the integrated J_{SC} extracted from the external quantum efficiency (EQE) data for the WBG (top cell) and NBG (bottom cell) subcells are 16.1 and 15.0 mA cm^{-2}, respectively (Figure 2c), which is in good agreement with the J_{SC} obtained from the $J-V$ measurements.

As illustrated in Figure 3a, cyclic tensile bending strain induces stress fatigue and residual-stress accumulation in the perovskite layer, which subsequently promotes defect formation and device degradation. To investigate the degradation behavior and the influence of bending-induced mechanical stress on 2T flexible PTSCs, GIXRD measurements were conducted. Tensile bending strain was applied by repeatedly folding the devices outward for 5000 cycles at a curvature radius of 5 mm. GIXRD was collected for the devices before and after 5000 bending cycles (Figure 3b−e). Structural information from both NBG and WBG perovskite layers in the PTSCs was simultaneously obtained by tuning the X-ray incident angle (ψ). The residual stress (σ_r) was then quantified using the 2θ-$\sin^2\psi$ method (Figure S5).[27,28] The analysis is based on the assumptions of linear elasticity, a bi axial in-plane stress state and quasi-isotropic behavior of the polycrystalline films. The peak shifts were attributed primarily to elastic lattice strain, with a uniform stress distribution within the films.[29] As illustrated in Figure S5a, an ψ was selected to allow X-ray penetration through both the NBG and WBG perovskite layers in the flexible tandem structure. Detailed measurement conditions are provided in the Experimental section. The σ_r is extracted from the slope of the linear $2\theta-\sin^2\psi$ relationship (see Figure S5b−e), which reflects the ψ-dependent lattice strain induced by in-plane stress, as derived from Bragg's law combined with the generalized Hooke's law in eq 2[27,28]

$$\sigma_r = -\frac{E\pi}{(1+\upsilon)180^\circ}\cot\theta_0\frac{\partial(2\theta)}{\partial(\sin^2\psi)} \quad (2)$$

where θ_0 is half of the scattering angle of the stress-free perovskite. Compared to the control perovskite films, the grain-boundary polymerization effectively reduces the initial σ_r from 19.47 ± 0.65 to 14.83 ± 0.70 MPa for NBG perovskite layer and from 35.24 ± 3.25 to 21.16 ± 1.62 MPa for WBG perovskite layer (Figure 3f,g). A comparable stress-relief effect is also observed in samples containing only NBG or WBG perovskite layers deposited on ITO/PEN flexible substrate (Figure S6). Compared to the single-junction results in Figure S6, the more pronounced residual stress evolution observed for the WBG layer in the tandem device (Figure 3g) indicates that cumulative strain introduced during fabrication of multilayers

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in the tandem architecture further amplifies the intrinsic mechanical sensitivity of the WBG composition. After 5000 bending cycles, the σ_r in the control NBG and WBG perovskite layers increases to 41.47 ± 1.86 and 61.33 ± 3.52 MPa, respectively. In contrast, target samples subjected to dual-layer in situ polymerization exhibit much smaller stress increases, reaching only 26.42 ± 1.40 and 38.37 ± 2.62 MPa. As shown in Figure S7, cross-sectional images of the flexible PTSCs before and after bending cycles reveal no visible signs of fracture or delamination. This observation indicates that the changes in σ_r arise from accumulation of tensile bending stress within the perovskite layer under bending conditions prior to the onset of macroscopic structural failure. The increase in residual stress indicates the accumulation of irreversible strain components during cyclic bending, which cannot be explained solely by reversible elastic deformation, and that the target process effectively mitigates the resulting irreversible lattice deformation.[26,30]

Figure 4a−c shows $J-V$ characteristics of the flexible single-junction NBG and WBG PSCs and the flexible 2T PTSCs before and after 5000 bending cycles for control and target devices. The corresponding changes in photovoltaic parameters are summarized in Figure S8. Upon bending stress applied to the control devices, J_{SC}, V_{OC}, and FF decrease from 30.64 to 28.89 mA cm^{-2} (decreasing rate: 5.7% (−5.7%)), from 0.717 to 0.706 V (−1.5%), and from 0.744 to 0.670 (−9.9%), respectively, for the single-junction NBG cell. This degradation is more pronounced in the single-junction WBG cell, where J_{SC} decreases from 17.06 to 14.97 mA cm^{-2} (−12.2%), V_{OC} from 1.213 to 1.190 V (−1.9%) and FF from 0.765 to 0.598 (−21.8%), which is probably due to higher E for WBG perovskite than for NBG one as confirmed in Figure 1. For the control PTSC, bending-induced degradation is also observed, with J_{SC} decreasing from 15.34 to 14.27 mA cm^{-2} (−7.0%), V_{OC} from 1.973 to 1.923 V (−2.5%), and FF from 0.779 to 0.703 (−9.8%). The bending-induced degradation of the PTSC correlates with the degradation behavior of both the NBG and WBG subcells, with the overall device performance being predominantly governed by the mechanically more vulnerable WBG subcell due to current matching and series-resistance effects. More pronounced degradation of the WBG single-junction device and WBG subcell can also be attributed to their intrinsically higher lattice rigidity arising from the Cs/Br-rich composition and strengthened Pb−X bonding, combined with the higher density of grain boundaries that can act as stress concentration sites.[31]

For target devices, the bending-induced degradation of photovoltaic parameters is substantially suppressed. In the single-junction NBG cell, J_{SC}, V_{OC} and FF decrease merely from 30.92 to 29.88 mA cm^{-2} (−3.4%), from 0.763 to 0.762 V (−0.1%), and from 0.747 to 0.704 (−5.8%), respectively. A similarly pronounced mitigation is observed in the single-junction WBG cell, where J_{SC}, V_{OC} and FF decrease slightly from 17.49 to 17.00 mA cm^{-2} (−2.8%), from 1.245 to 1.237 V (−0.6%), and from 0.815 to 0.725 (−11.0%). As a result, the bending-induced degradation of the photovoltaic parameters in the dual-layer polymerized PTSC is effectively mitigated, resulting in only a slight decrease in J_{SC} from 15.34 to 14.79 mA cm^{-2} (−3.6%), V_{OC} from 2.008 to 1.998 V (−0.5%), and FF from 0.791 to 0.750 (−5.2%). In contrast to the control PTSC, which exhibits pronounced bending-induced degradation, the dual-layer polymerized PTSC shows markedly improved mechanical robustness. Consequently, the PCE of the control PTSC is significantly reduced from 23.56% to 19.29% (−18.1%), whereas the target PTSC shows an approximately 2-fold mitigation of performance loss, with the PCE decreasing only from 24.34% to 22.15% (−9.0%). Detailed photovoltaic parameters before and after the bending stability test are listed in Table S4. MPPT measurements further show that the target flexible PTSC maintains a stable output after bending cycles, whereas the control device exhibits gradual performance degradation, indicating preserved operational stability against mechanical stress by target process (Figure S9). Additional bending tests were conducted under more severe bending conditions with radii below 5 mm (Figure S10). After sequential bending tests at r = 5, 4, 3, and 2 mm, the target device consistently exhibited higher PCE retention than the control device despite the rapid degradation observed under these extreme mechanical deformation conditions. These results further support the effectiveness of the target treatment in mitigating mechanically induced degradation under high tensile-stress conditions. Among previously reported flexible PTSCs, our target flexible PTSC achieves both high efficiency and outstanding mechanical durability under severe bending conditions (Table S5). In particular, the device retains over 90% of its initial PCE after 5000 bending cycles at a bending radius of 5 mm, which represents one of the most stringent bending conditions reported for flexible PTSCs. In addition, the dual-layer grain-boundary polymerization strategy shows broad applicability to both WBG and NBG subcells, whereas many previously reported strategies have mainly focused on individual subcells.

Since the bending-induced degradation is most pronounced in the single-junction WBG cell and consequently governs the degradation behavior of the PTSC, TRPL analyses were performed to investigate the influence of bending strain on nonradiative recombination and charge transfer in the WBG perovskite and grain-boundary polymerization effect as well. TRPL decay of the WBG perovskite layers with and without electron transport layer (ETL) was investigated before and after repetitive bending strain (Figure 4d,e). The time-dependent PL decay data were analyzed by fitting to a biexponential decay model as listed in Table S6.[32] For the WBG perovskite deposited on PEN substrate, the average carrier lifetime (τ_{avg}) decreases from 560 to 325 ns in the control sample after 5000 bending cycles, whereas a smaller reduction from 784 to 509 ns is observed in the target sample. This indicates that grain-boundary polymerization suppresses the formation of the bending-induced defects. For the ETL(BCP/C_{60})/perovskite/PEN flexible samples, τ_{avg} increases from 3.25 to 8.80 ns after 5000 bending cycles, whereas in the target sample τ_{avg} changed from 2.84 to 4.22 ns. The shorter τ_{avg} in the presence of ETL results from the charge extraction and the increased τ_{avg} is indicative of poor charge extraction due to repetitive bending. Compared with the control samples, the reduced τ_{avg} values in the target samples both before and after bending are indicative of the stabilizing effect of grain-boundary polymerization on the charge transfer system under mechanical deformation.[33] EIS was studied with the flexible WBG single junction before and after bending strain to elucidate its impact on heterojunctional properties. Based on the equivalent circuit model presented in Figure S11, the series resistance (R_s), charge transport resistance (R_{ct}), and recombination resistance (R_{rec}) are extracted from Nyquist plots (Table S7).[34] The incorporation of grain-boundary polymerization inherently leads to lower R_s and R_{ct} along with

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higher R_{rec} than those of the control sample, which indicates that grain-boundary polymerization improves charge extraction, enhances charge transport interface quality and suppresses nonradiative carrier recombination. Bending increases R_s and R_{ct} while decreasing R_{rec} in both the control and target devices. The relative change (%) was calculated using $(R_f - R_i)/R_i \times 100\%$, where R_i is the initial resistance without bending and R_f is the resistance after 5000 bending cycles; the results are plotted in Figure 4f. A pronounced difference in the relative change of R_s is observed between the control and target samples with corresponding increase in $J-V$ slope in near V_{oc} (Figure 4b), while the target devices exhibiting a much smaller variation. This reduced change accounts for the improved mechanical stability upon bending without significant deterioration in J_{SC}, FF and hysteresis,[35] as shown in Figure 4b and Table S4. The results also suggest that interfacial charge-transport and recombination processes are also affected during bending, indicating that the degradation behavior involves coupled bulk and interfacial contributions in addition to stress accumulation within the perovskite layer.

4. CONCLUSION

We demonstrated that the dual-layer grain-boundary polymerization in both NBG and WBG perovskite subcells markedly enhanced the mechanical durability of flexible 2T PTSCs by reducing the elastic modulus and thereby mitigating strain-induced stress. In addition to improved mechanical resilience, the in situ polymerization effectively passivated grain-boundary defects, leading to enhanced photovoltaic performance. Cyclic bending tests conducted at a radius of 5 mm for 5000 cycles revealed that grain-boundary polymerization in both NBG and WBG perovskite layers suppressed the accumulation of residual stress under repeated deformation. Notably, the WBG perovskite exhibited greater sensitivity to mechanical strain than the NBG counterpart, indicating that the mechanical stability of the WBG subcell largely governed the degradation behavior of flexible PTSCs. These results highlighted that grain-boundary polymerization across both perovskite layers was essential for achieving mechanically robust and high-performance flexible tandem devices.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.6c05001.

> Schematics of in situ polymerization reaction between DGEBA and IPDA. Statistical photovoltaic parameters of NBG and WBG perovskite single-junction cells. Load−displacement curves of perovskite films on rigid substrates. GIXRD patterns and $2\theta-\sin^2\psi$ plots for residual stress analysis. Cross-sectional SEM images before and after bending cycles. Photovoltaic parameter variations under bending stress. EIS Nyquist plots of WBG single junction cells and equivalent circuit analysis. TRPL fitting parameters for WBG samples. Photovoltaic parameters of the best performing cells and cells before and after bending cycles. Comparison of bending durability with previously reported flexible perovskite tandem solar cells (PDF)

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Author Contributions

#Y.-W.C. and K.-W.Y. These authors contributed equally to this work. Y.-W.C., K.-W.Y. and N.-G.P. conceived the idea. H.S. and N.-G.P. supervised the project. Y.-W.C., K.-W.Y. carried out the experiments, analyzed the data and wrote the manuscript. N.-G.P. edited the manuscript. H.S., J.H.N. and N.-G.P. provided research guidance. U.A. performed ALD-SnO_2 deposition for the tandem structures. G.S.Y. measured and analyzed EQE data of the tandem devices. H.-J.J. contributed to the tandem device fabrication. All authors reviewed the manuscript.

Notes

The authors declare no competing financial interest.

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